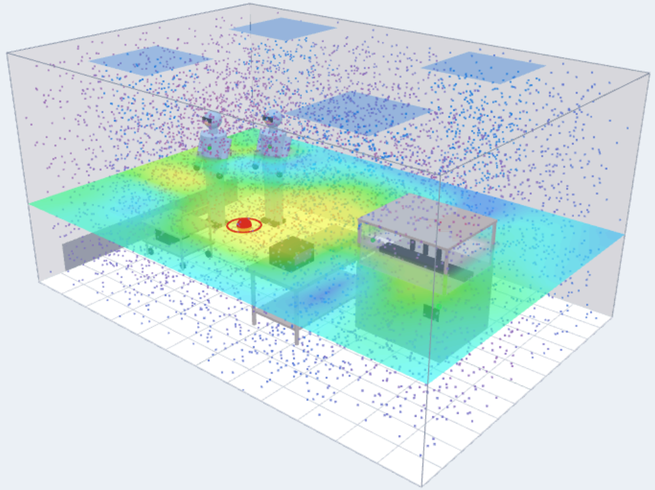
Flowcast

Cut ventilation energy by up to half.

A neural CFD surrogate predicts airflow in real time on a digital twin of your room, powered by AMD. Trained on 7,500 OpenFOAM simulations; recommendations confirmed by real CFD.



AMD Radeon + ROCm | Gemma 4 on AMD Token Factory | OpenFOAM | Open source, MIT

Martin Omwenga · AMD Developer Hackathon, Intelligent Industry

THE PROBLEM

Fans run at worst case, all day, because predicting a lower setting takes a specialist and a CFD study.

- Cleanrooms, labs and fan-cooled cabinets keep fans at one fixed, safe-by-assumption setting.
- A flow simulation could predict a lower one, but each run needs **a specialist and hours of computing**.
- So it is done once, at design time, and **never again** as the room changes.

~27%

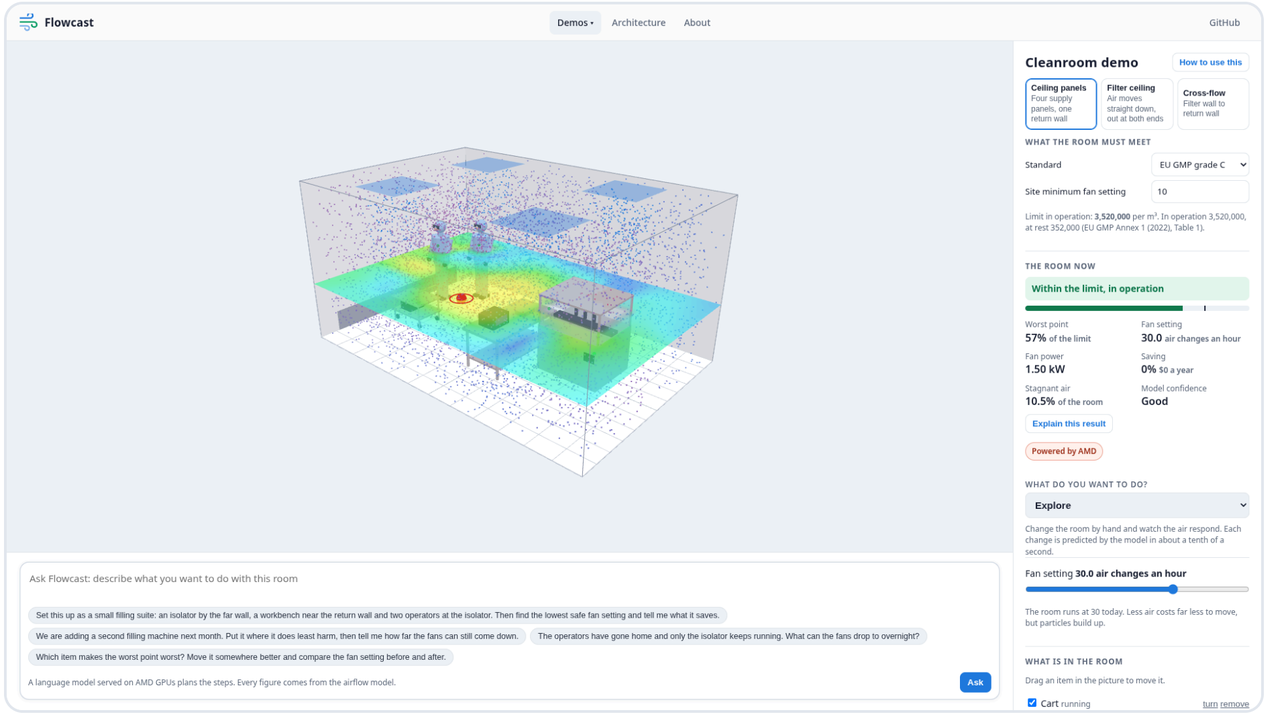
less fan power from 10% less air

~49%

less fan power from 20% less air: power follows the cube of airflow

THE SOLUTION

Simulate once in bulk. Then anyone can ask, in a tenth of a second.



- **We simulated** each room design a few thousand times in OpenFOAM.
- **Two 3D neural networks** learn the airflow and each source's particles or heat.
- **The person running the room** drags machines and people, and sees the airflow, the worst point and the verdict against a named standard.

WHAT IT DOES

From "what if?" to a confirmed fan setting.

- **Predict** the airflow, particles or heat as you drag: about 0.1 s on an AMD Radeon GPU.
- **Find the lowest safe setting** against EU GMP Annex 1 grades, ISO 14644-1 classes or a component rating.
- **Confirm by simulation:** a real OpenFOAM run checks the answer and steps the fan up until one passes.
- **Find the best place** for a machine or a component.
- **Ask in plain words:** Gemma 4 plans the steps; every number comes from Flowcast's own tools.
- **Plan a day:** paste a production plan; get a fan schedule with lead and clean-up periods.
- **Two demos:** a cleanroom in three ventilation designs, and a fan-cooled control cabinet.

HOW IT WORKS

Physics in bulk, then a fast model, then a check.

1. Simulate

7,500 rooms in OpenFOAM: steady airflow, one tracer per source, age of air.

2. Learn

A flow U-Net and a particle U-Net, 2 to 4 per design; their disagreement is the confidence reading.

3. Answer

Fields add up per source, so switching a machine off is instant; searches run hundreds of predictions.

4. Confirm

The best answer goes back to OpenFOAM; the fan steps up until a real simulation passes.

AMD Radeon GPU + ROCm
Trains and serves the airflow model with PyTorch.

AMD Instinct via Token Factory
Serves Gemma 4, the plain-language agent, with DeepSeek as fallback.

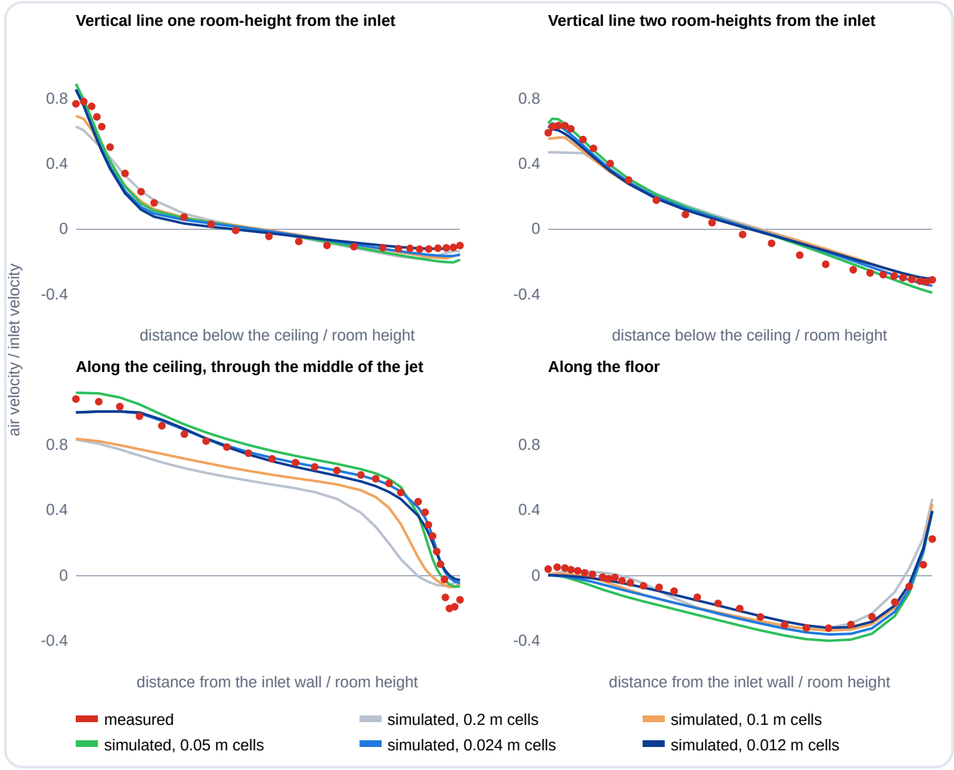
AMD EPYC in the cloud
Hosts the public demo, around the clock.

THE EVIDENCE

Every number is measured.

On rooms it never saw	Same verdict as simulation	Worst point error (median)
Cleanroom, panels	92 to 98%	9 to 13%
Filter ceiling	98%	4%
Cross-flow	94%	6%
Control cabinet	94%	10 to 13%

Ranges: random and realistic layouts. Realistic rooms were twice as hard at first; 1,000 realistic rooms added to training closed most of the gap.



Checked against a real experiment: the IEA Annex 20 room, laser-measured air velocity. Our simulation set-up matches within **4.3%** on a fine grid.

MARKET

Proven demand: industry already pays to cut cleanroom air.

- **Who:** pharmaceutical and biotech manufacturing, electronics, hospital pharmacies and labs; then panel builders and data rooms for the heat version.
- **Why now:** energy costs, and ISO 14644-16, a standard on cleanroom energy efficiency.
- **Evidence of demand:** Roche/Genentech runs an air-change-reduction programme; vendors sell airflow reduction.

60–80%

fan energy cut by demand-controlled filtration (Lawrence Berkeley National Laboratory)

Every change

a new layout, machine or schedule is a new question that today goes unanswered

Sources: LBNL, ISPE (Roche/Genentech), ISO 14644-16

COMPETITION

Others react or study once. Flowcast predicts before a change.

Approach	How it works	Limit
Reactive control (EECO2, Telstar SmartCR)	Particle counters drive the fan speed	Sees only where counters are; finds out after the fact; needs hardware and a retrofit
Consultant CFD study	One simulation study at design time	Weeks and a specialist per question; never repeated
AI twin for chip fabs (TwinFab)	Physics plus AI for semiconductor fabs	No published accuracy; aimed at the largest sites
Flowcast	**Predicts the whole room, any change, in a tenth of a second; confirms by simulation**	**Published error, open source, aimed at ordinary cleanrooms and cabinets; works alongside controllers**

BUSINESS MODEL AND NEXT STEPS

Describe your room. Get its airflow.

- **The goal:** describe a room in plain words or upload a drawing; Gemma builds its digital twin and our model predicts the airflow. Today that works within a growing library of trained room designs, and every design we simulate widens it.
- **Subscription per room:** daily use for change planning, night setback and fan schedules; new room types added to the library on demand.
- **Sold through** cleanroom builders, HVAC service firms and validation consultants, who gain a faster study.
- **Open core:** the code is open; set-up, hosting

- **Next:** more room types; a finer training grid, the largest remaining error.
- People who move: a heatmap of where they spend their time.
- Learn from use: every confirmed room becomes training data.
- Live readings: compare predictions with the room's own sensors.



Flowcast

Try it.

Live demo: https://100-31-84-106.sslip.io
Code: github.com/MrtnOmwenga/flowcast